Task 1

```
import os
import cv2
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras.layers import Conv2D, MaxPooling2D, UpSampling2D, Input
from tensorflow.keras.models import Model
```

```
dataset_path = Path("BrainMRI/yes")

image = list(dataset_path.glob("*.jpg"))[0]
print("Image Path:", image)
```

```
Image Path: BrainMRI/yes/Y186.jpg
```

```
image = cv2.imread(image)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
gray = cv2.cvtColor(image, cv2.COLOR_RGB2GRAY)
gray = cv2.resize(gray, (128, 128))
gray = gray.astype(np.float32) / 255.0
```

```
input_image = np.expand_dims(gray, axis=-1)
input_image = np.expand_dims(input_image, axis=0)

print(input_image.shape)
```

```
(1, 128, 128, 1)
```

Task 2

```
inputs = Input(shape=(128,128,1))

# Encoder
c1 = Conv2D(16, (3,3), activation='relu', padding='same')(inputs)
p1 = MaxPooling2D((2,2))(c1)

# Bottleneck
c2 = Conv2D(32, (3,3), activation='relu', padding='same')(p1)

# Decoder
u1 = UpSampling2D((2,2))(c2)
c3 = Conv2D(16, (3,3), activation='relu', padding='same')(u1)
```

```
# Output layer
outputs = Conv2D(1, (1,1), activation='sigmoid')(c3)

model = Model(inputs, outputs)

model.summary()
```

Model: "functional_6"

Layer (type)	Output Shape	Param #
input_layer_7 (InputLayer)	(None, 128, 128, 1)	0
conv2d_4 (Conv2D)	(None, 128, 128, 16)	160
max_pooling2d_1 (MaxPooling2D)	(None, 64, 64, 16)	0
conv2d_5 (Conv2D)	(None, 64, 64, 32)	4,640
up_sampling2d_1 (UpSampling2D)	(None, 128, 128, 32)	0
conv2d_6 (Conv2D)	(None, 128, 128, 16)	4,624
conv2d_7 (Conv2D)	(None, 128, 128, 1)	17

Total params: 9,441 (36.88 KB)
Trainable params: 9,441 (36.88 KB)

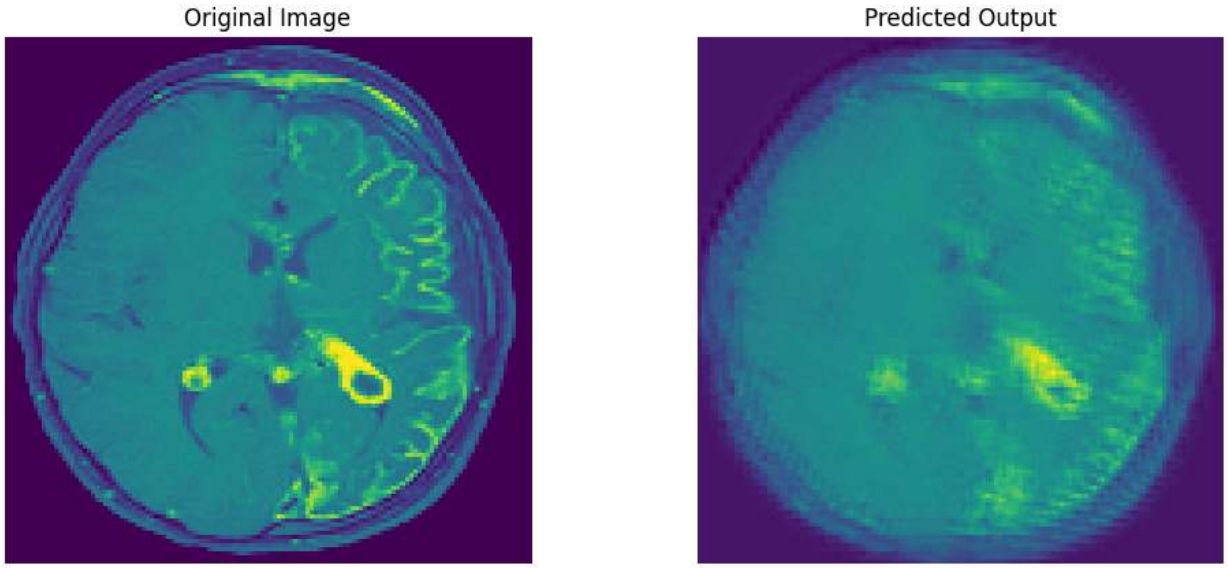
```
prediction = model.predict(input_image)
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.imshow(gray)
plt.title("Original Image")
plt.axis('off')

plt.subplot(1,2,2)
plt.imshow(prediction[0,:,:,0])
plt.title("Predicted Output")
plt.axis('off')

plt.show()
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 151ms/step
```



Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.